

Experiment 10

Dockerfile is a text file that contains a set of instructions used to automatically build a Docker image. It defines the base image, environment, dependencies, and commands required to run an application inside a container.

1. Dockerfile Overview

A Dockerfile consists of step-by-step instructions that Docker follows to create an image. Each instruction in the Dockerfile creates a new layer in the image.

Common Dockerfile instructions include:

- **FROM:** Specifies the base image
- **RUN:** Executes commands during image build
- **COPY / ADD:** Copies files into the container
- **WORKDIR:** Sets working directory
- **CMD / ENTRYPOINT:** Defines default command to run

2. Pulling an Image from Docker Hub

The command:

```
docker pull hello-world
```

is used to download an image from Docker Hub.

Explanation:

- Downloads the hello-world image from the official Docker repository
- If not present locally, Docker fetches it from the remote registry
- The image is stored locally for future use
- Output shows layers being downloaded and image digest

3. Listing Available Docker Images

The command:

```
docker images
```

displays all locally available

Docker images.

Explanation:

- Shows repository name, tag, image ID, size, and creation details
- Confirms successful image download
- Helps manage and identify images stored in the system

4. Building a Docker Image

A Docker image is built using the command:

```
docker build -t <image_name> .
```

Explanation:

- Builds an image from the Dockerfile in the current directory
- -t is used to assign a name (tag) to the image
- Docker processes each instruction in the Dockerfile step-by-step
- Final output is a reusable image

5. Running a Docker Container

The command:

docker run hello-world is used to run a container from the specified image.

Explanation:

- Docker checks if the image is available locally
- If not, it pulls the image from Docker Hub
- Creates a new container from the image
- Executes the default command defined in the image
- Displays output in the terminal

6. Working of Hello-World Container

When the hello-world container is executed, the following steps occur:

- Docker client sends request to Docker daemon
- Docker daemon pulls the image (if not available locally)
- A container is created from the image
- The container runs a simple program that prints a message
- Output is sent back to the terminal

This confirms that Docker is installed and functioning correctly.

7. Image Storage and Layers

Docker images are built using a layered architecture:

- Each instruction in a Dockerfile creates a layer
- Layers are cached to improve performance
- Images are stored in the local Docker registry

Conclusion

Thus, Dockerfile provides a structured way to define application environments and build Docker images. Commands like `docker pull`, `docker images`, `docker build`, and `docker run` help in managing images and running containers. The successful execution of the `hello-world` container confirms proper Docker setup and understanding of image execution workflow.

```
C:\Users\15L>docker pull hello-world
Using default tag: latest
latest: Pulling from library/hello-world
17eec7bbc9d7: Pull complete
Digest: sha256:ef54e839ef541993b4e87f25e752f7cf4238fa55f017957c2eb44077083d7a6a
Status: Downloaded newer image for hello-world:latest
docker.io/library/hello-world:latest

What's next:
  View a summary of image vulnerabilities and recommendations → docker scout quickview hello-world

C:\Users\15L>|
```

```
C:\Users\15L>docker images

IMAGE                ID                DISK USAGE    CONTENT SIZE    EXTRA
hello-world:latest   1b44b5a3e06a      10.1kB        0B
httpd:latest         4613a77dcb46      117MB         0B              U
mysql:latest         a88c3e85e887      632MB         0B
nginx:1.0            fc bd66995946      169MB         0B              U
node:latest          c3978d05bc68      1.1GB         0B              U
redis:latest         170a1e90f843      138MB         0B
ubuntu:latest        ca2b0f26964c      77.9MB        0B              U

C:\Users\15L>|
```

```
C:\Users\15L>docker run hello-world
```

```
Hello from Docker!
```

```
This message shows that your installation appears to be working correctly.
```

To generate this message, Docker took the following steps:

1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
(amd64)
3. The Docker daemon created a new container from that image which runs the executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it to your terminal.

To try something more ambitious, you can run an Ubuntu container with:

```
$ docker run -it ubuntu bash
```

Share images, automate workflows, and more with a free Docker ID:

<https://hub.docker.com/>

For more examples and ideas, visit:

<https://docs.docker.com/get-started/>

```
C:\Users\15L>
```

```
C:\Users\15L>docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
f69ff2c3de9b	hello-world	"hello"	About a minute ago	Exited (0) About a minute ago		quizzical_bell
0d8d880ac2d2	httpd	"httpd-foreground"	4 months ago	Exited (0) 4 months ago		flamboyant_knuth
e81e8ac6c5c6	httpd	"httpd-foreground"	4 months ago	Exited (0) 4 months ago		jolly_herschel
f99bf8580ef9	httpd	"httpd-foreground"	4 months ago	Exited (0) 4 months ago		inspiring_montalcini
c1558a5bdfc7	httpd	"httpd-foreground"	4 months ago	Exited (0) 4 months ago		compassionate_cerf
61c392b4db8a	httpd	"httpd-foreground"	4 months ago	Exited (0) 4 months ago		suspicious_heisenberg
a916b1bcd1d3	httpd	"httpd-foreground"	4 months ago	Exited (0) 4 months ago		nice_roentgen
004bcde789ff	httpd	"httpd-foreground"	4 months ago	Exited (0) 4 months ago		sad_noyce
c8cb91659812	httpd	"httpd-foreground"	4 months ago	Exited (0) 4 months ago		flamboyant_pike
d69547f1917e	httpd	"httpd-foreground"	4 months ago	Exited (0) 4 months ago		intelligent_greider
< 73a0452c	httpd	"httpd-foreground"	4 months ago	Exited (0) 4 months ago		zealous_bouman
< 08ef6efe	httpd	"httpd-foreground"	4 months ago	Exited (0) 4 months ago		sad_elion
74b68ec74672	httpd	"httpd-foreground"	4 months ago	Exited (0) 4 months ago		hungry_jang
55a80be3d445	httpd	"httpd-foreground"	4 months ago	Exited (0) 4 months ago		youthful_matsumoto
4ed8cfaded6b	httpd	"httpd-foreground"	4 months ago	Exited (0) 4 months ago		sleepy_gates
2fa633478597	ubuntu	"/bin/bash"	23 months ago	Exited (255) 6 months ago		unruffled_wilson
ee37479b71cd	node	"docker-entrypoint.s..."	24 months ago	Exited (0) 23 months ago		sarth
9e7ea042694c	node	"docker-entrypoint.s..."	24 months ago	Exited (0) 23 months ago		lucid_faraday
f4b59b6cf1f4	d1397258b209	"docker-entrypoint.s..."	24 months ago	Exited (255) 6 months ago	6379/tcp	musing_archimedes
d82b6fa1a86	d2c94e258dcb	"hello"	24 months ago	Exited (0) 24 months ago		bb71ba51783
bb71b1a51783	d2c94e258dcb	"hello"	24 months ago	Exited (0) 24 months ago		silly_wozniak
922a7dec7144	d1397258b209	"docker-entrypoint.s..."	24 months ago	Exited (255) 24 months ago	6379/tcp	festive_hertz
97ba06c14c7	d2c94e258dcb	"hello"	24 months ago	Exited (0) 24 months ago		b9f4789574a5
ae743b95da80	d2c94e258dcb	"hello"	24 months ago	Exited (0) 24 months ago		mystifying_ramanujan
b9f4789574a5	d2c94e258dcb	"hello"	24 months ago	Exited (0) 24 months ago		interesting_yalow
a21892131937	nginx:1.0	"/usr/sbin/nginx -g ..."	1 years ago	Exited (255) 24 months ago	0.0.0.0:9090->80/tcp	webserver